

Assignment8

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```
install.packages("fivethirtyeight")
install.packages("tidyverse")
install.packages("scales")
library(fivethirtyeight)
library(tidyverse)
library(scales)

data("bechdel")

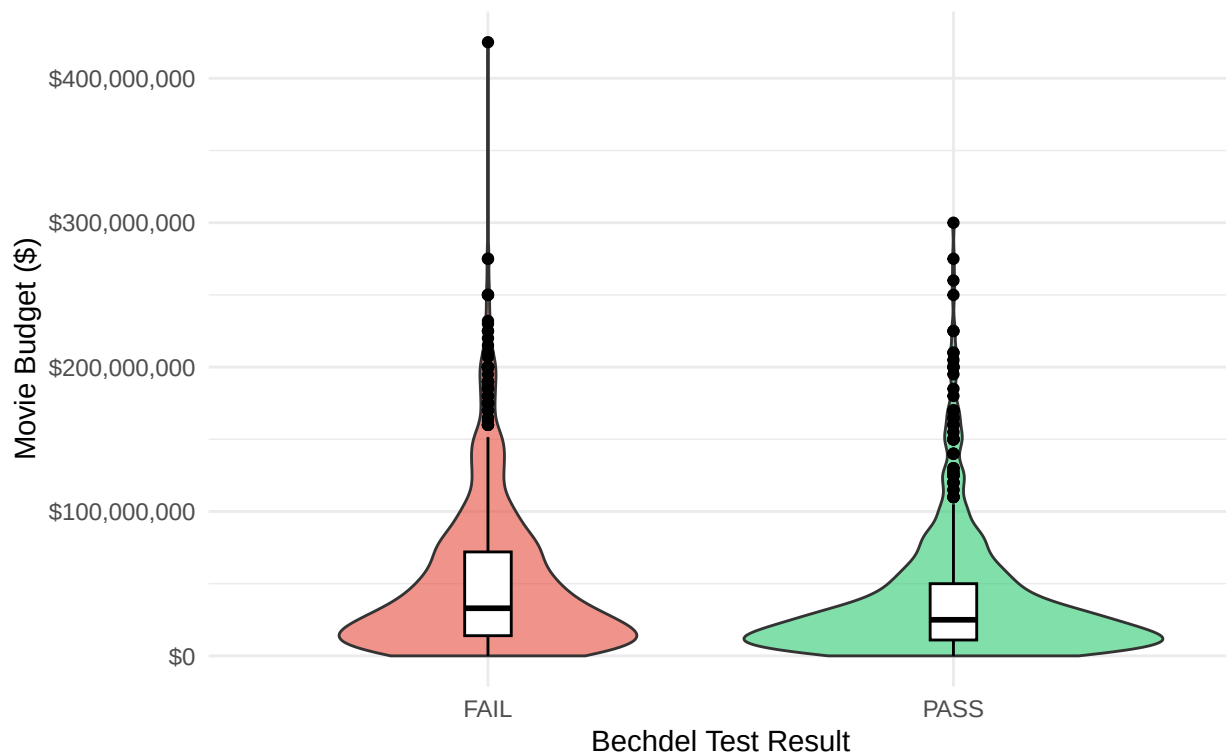
df_hollywood <- bechdel %>%
  filter(!is.na(budget) & !is.na(domgross)) %>%
  mutate(test_result = ifelse(clean_test == "ok", "PASS", "FAIL")) %>%

  mutate(roi = (domgross - budget) / budget) %>%
  mutate(decade = floor(year / 5) * 5)

ggplot(df_hollywood, aes(x = test_result, y = budget, fill = test_result)) +
  geom_violin(alpha = 0.6) + # Dağılımın şeklini gösterir
  geom_boxplot(width = 0.1, color = "black", fill = "white") +
  scale_y_continuous(labels = dollar_format()) +
  scale_fill_manual(values = c("FAIL" = "#E74C3C", "PASS" = "#2ECC71")) +
  labs(title = "The Budget Gap",
       subtitle = "Male-dominated movies receive significantly higher budgets",
       x = "Bechdel Test Result",
       y = "Movie Budget ($)") +
  theme_minimal() +
  theme(legend.position = "none")
```

The Budget Gap

Male-dominated movies receive significantly higher budgets



Movies that pass the test usually stop around \$200 or \$300 million. However, movies focused on men can reach \$400 million. This shows that Hollywood gives the biggest budgets and takes the biggest risks mostly for male stories. Both distributions are widest at the bottom, suggesting that low-to-mid budget films are common in both categories. However, the wider upper body of the red violin shows that mid-high budget tier is also dominated by movies that fail the test.

```
movies <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday/master/data/2020/02/movies.csv')
```

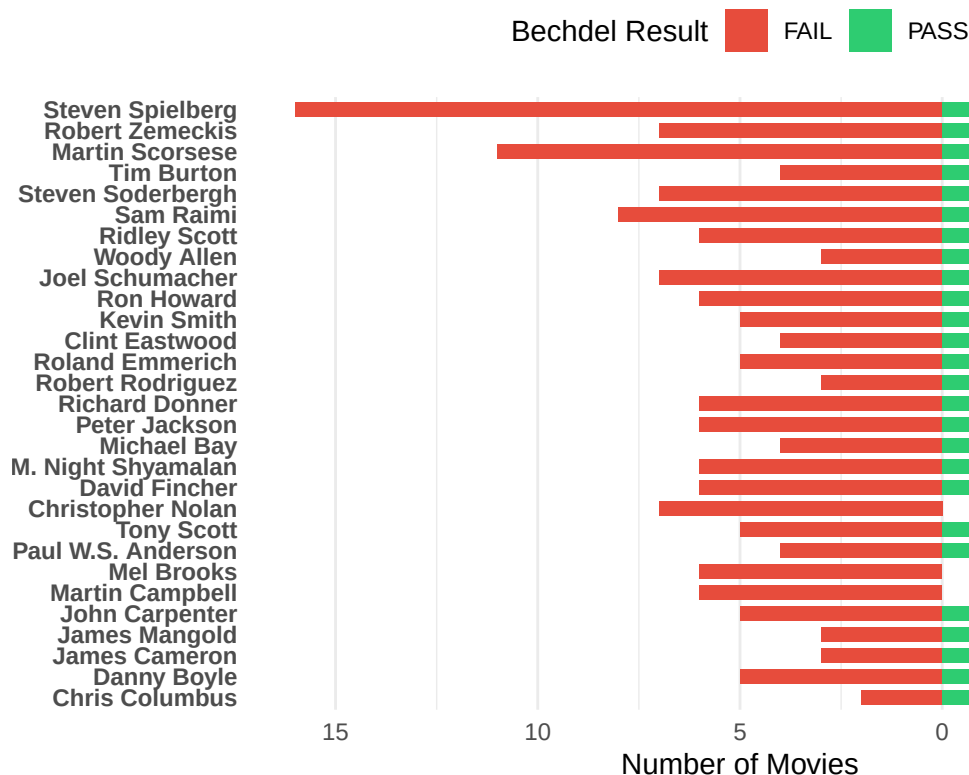
```
director_df <- movies %>%  
  filter(!is.na(director)) %>%  
  mutate(test_result = ifelse(clean_test == "ok", "PASS", "FAIL")) %>%
```

```
  group_by(director) %>%  
  filter(n() >= 5) %>%  
  ungroup()
```

```
director_stats <- director_df %>%  
  group_by(director, test_result) %>%  
  summarise(count = n(), .groups = 'drop') %>%  
  
  mutate(plot_count = ifelse(test_result == "FAIL", -count, count)) %>%  
  
  group_by(director) %>%  
  mutate(total_films = sum(count)) %>%  
  ungroup() %>%  
  
  slice_max(total_films, n = 40)
```

```
ggplot(director_stats, aes(x = reorder(director, total_films), y = plot_count, fill = test_result)) +
  geom_col(width = 0.7) +
  coord_flip() +
  scale_y_continuous(labels = abs) +
  scale_fill_manual(values = c("FAIL" = "#E74C3C", "PASS" = "#2ECC71")) +
  labs(title = "Directors' Equality Scorecards",
       x = NULL,
       y = "Number of Movies",
       fill = "Bechdel Result") +
  theme_minimal() +
  theme(
    legend.position = "top",
    panel.grid.major.y = element_blank(),
    axis.text.y = element_text(size = 9, face = "bold")
  )
```

Directors' Equality Scorecards



Steven Spielberg and Martin Scorsese have very long red bars on the left. This shows that although they make popular movies, their stories are almost entirely focused on men. Some directors show a complete lack of female narratives. For example, looking at Christopher Nolan, we see a red bar but almost no green bar. This suggests that his movies consistently ignore female relationships. On the other hand, directors like Tim Burton and Woody Allen have longer green bars. This proves that it is possible to be a famous director and still tell stories that include women meaningfully.

```
library(tidyverse)

sunburst_data <- movies %>%
  filter(!is.na(clean_test)) %>%
  mutate(
```

```

main_status = ifelse(clean_test == "ok", "PASS", "FAIL"),

detail_status = case_when(
  clean_test == "ok" ~ "Passed",
  clean_test == "men" ~ "Talked about Men",
  clean_test == "notalk" ~ "No Dialogue",
  clean_test == "nowomen" ~ "No Women",
  clean_test == "dubious" ~ "Dubious",
  TRUE ~ "Other"
)
)

outer_ring <- sunburst_data %>%
  count(main_status, detail_status) %>%
  arrange(main_status, desc(detail_status)) %>%
  mutate(
    fraction = n / sum(n),
    ymax = cumsum(fraction),
    ymin = c(0, head(ymax, n = -1))
  )

inner_ring <- sunburst_data %>%
  count(main_status) %>%
  arrange(main_status) %>%
  mutate(
    fraction = n / sum(n),
    ymax = cumsum(fraction),
    ymin = c(0, head(ymax, n = -1))
  )

ggplot() +
  geom_rect(data = outer_ring,
    aes(ymin = ymin, ymax = ymax, xmin = 3, xmax = 4, fill = detail_status),
    colour = "white") +

  geom_rect(data = inner_ring,
    aes(ymin = ymin, ymax = ymax, xmin = 0, xmax = 3, fill = main_status),
    colour = "white") +

  coord_polar(theta = "y") +

  scale_fill_manual(values = c(
    "PASS" = "#2ECC71",
    "Passed" = "#2ECC71",
    "FAIL" = "#E74C3C",
    "Talked about Men" = "#C0392B",
    "No Dialogue" = "#E59866",
    "No Women" = "#D35400",

```

```

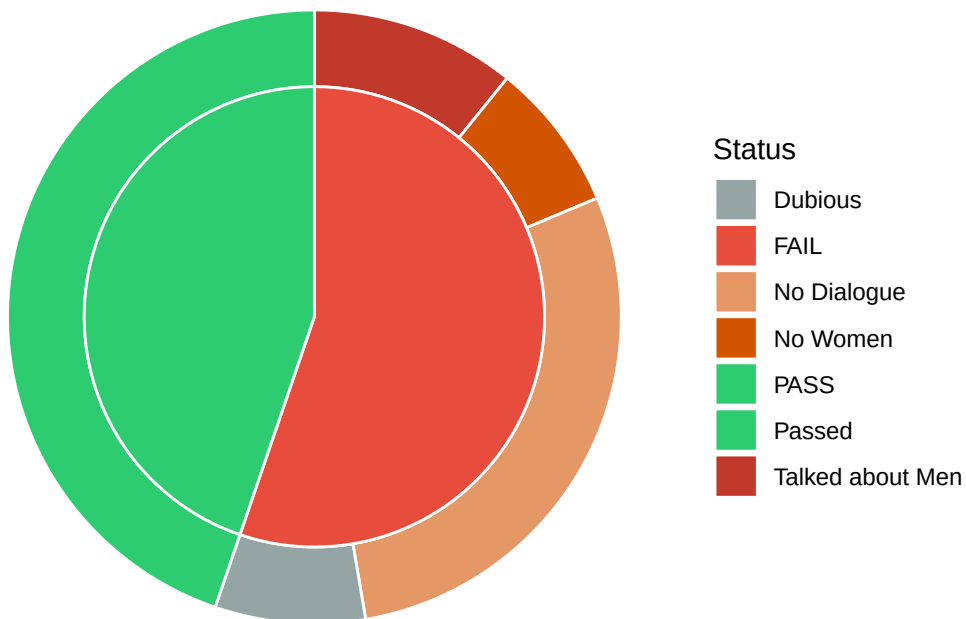
"Dubious" = "#95A5A6",
"Other" = "grey"
)) +

theme_void() +
labs(title = "The Anatomy of Failure",
      subtitle = "Inner Ring: General Status | Outer Ring: Why Failed?",
      fill = "Status") +
theme(plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
      plot.subtitle = element_text(hjust = 0.5))

```

The Anatomy of Failure

Inner Ring: General Status | Outer Ring: Why Failed?



The biggest slice is 'Talked about Men'. It means women are present in the movie, and they speak to each other, but their topic is always a man. The smallest slice in the failure section is 'No Women'. Women are on the screen, but the scripts do not give them independent stories or conversations.